CSC343H5 S

Winter 2026 Syllabus

Course Meetings

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Section	Day & Time	Delivery Mode & Location
LEC0104	Wednesday, 9:00 AM - 11:00 AM	In Person: MN 2170
LEC0103	Wednesday, 3:00 PM - 5:00 PM	In Person: MN 2170
LEC0102	Wednesday, 11:00 AM - 1:00 PM	In Person: MN 2170
LEC0101	Tuesday, 11:00 AM - 1:00 PM	In Person: MN 2170
PRA0101	Monday, 10:00 AM - 11:00 AM	In Person: DH 2020
PRA0102	Monday, 10:00 AM - 11:00 AM	In Person: DH 2026
PRA0103	Monday, 11:00 AM - 12:00 PM	In Person: DH 2020
PRA0104	Monday, 12:00 PM - 1:00 PM	In Person: DH 2020
PRA0105	Monday, 1:00 PM - 2:00 PM	In Person: DH 2026
PRA0106	Monday, 1:00 PM - 2:00 PM	In Person: DH 2020
PRA0107	Monday, 2:00 PM - 3:00 PM	In Person: DH 2020
PRA0108	Monday, 2:00 PM - 3:00 PM	In Person: DH 2026

Refer to ACORN for the most up-to-date information about the location of the course meetings.

Overview of Topics

1. Introduction to Databases
2. The Relational Model
3. The Entity-Relationship Model
4. SQL (introduction, DDL, DML, constraints, triggers)
5. Relational Algebra
6. Views and Indexes
7. Relational Database Design (Design Theory, Schema Decomposition, Normalization)
8. Transactions and Concurrency Control (Properties, Conflicts, Recoverability, Locking)

Course Website/Online Content

Quercus (and the MCS Download Site) will be used to host all of this course's online content (e.g., slides, exercises, handouts, assignments, etc.).

MarkUs will be used for the collection of student's submissions for assessed work completed outside a supervised setting (e.g., assignments).

PCRS will be used to host asynchronous content and exercises.

Piazza will be used as the course Q&A/Discussion board. This will also be used for course announcements.

Other Educational Technologies like VoiceEx, QuickTA, Microsoft/Google Forms, other custom systems, etc. may be used to enhance active problem solving, learning, or other exercises. These systems may be integrated/linked from Quercus, PCRS, Assignment PDFs, etc.

Lectures

Lectures start on Week 1 for this course and will occur in-person. Lectures will **NOT** be recorded, these are meant for you to actively participate in and problem solve synchronously in-person with your classmates and the lecturer.

Practicals

Practicals start on Week 2 for this course and will occur in-person. Practicals will **NOT** be recorded, these are meant to support your database skills and ability to complete the assignments. Practicals will contain active components to participate in and require you to problem solve synchronously in-person with your classmates and the Teaching Assistant.

Piazza (Q&A / Discussion Board)

Students are expected to check Piazza daily.

Please do not email Course Instructors content-specific questions, this course will be utilizing Piazza for Q&A. Students, Teaching Assistants, and Instructor(s) have the opportunity to answer questions. If you believe that the majority of students would benefit to an answer, please make a public Piazza post. If your question divulges a solution or is one which should not be publicly available, please make a private Piazza post.

Students found misbehaving or acting inappropriately on the discussion board may have their account suspended or permanently banned. This will be left to the discretion of the Course Instructor(s).

Note: if the question is of a personal/private nature, send an email.

Email Policy

Emails will *usually* be returned within three (3) business days (i.e., 72-hours, not including weekends and school recognized holidays/breaks). Emails are reserved for personal matters, content related questions should be directed to Piazza.

Office Hours

Office Hours take place in the Course Instructor's office. The location is provided under Contact Information on the syllabus, but additional details will be provided on Quercus.

Remember, you also have access to Piazza which often yields a faster reply and makes for a good resource.

Assignments

The assignments in this course relies on your ability to read, parse, and interpret large amounts of data/information. At times, this information may be overwhelming and ambiguous. As computer scientists, it is crucial that you turn this ambiguity into something useful to solve your assignments. To assist with this, make sure to regularly check and utilize Piazza to seek clarification and to collaborate on parsing and understanding assignment details in your practical sections with your Teaching Assistants and Classmates. Remember, it is okay to discuss strategies and seek clarification on assignments as a class, however, you must not directly share full or partial solutions. The work you submit is your work and your work alone.

Re-Mark/Re-Grade Requests

Re-Mark requests must be submitted to a member of the Teaching Team (on MarkUs) **within 5 calendar days of receiving a grade**. Re-Mark requests after this time will not be accepted.

Minimum Standards for Submitted Work

For your assessments/submissions (e.g., Assignments) to be graded, it must meet the minimum standards of a professional computer scientist. All files required to build the program must be submitted, and the program must compile cleanly, without errors or warnings on the MCS lab machines. Scripts and/or SQL files should be runable without error (note: commenting out all the code in a file to make it executable is not permissable, files should run without alteration).

Last minute difficulties with version control can easily be avoided by ensuring all files are added to the repository well before the deadline, and that you know how to commit and push them. You can and should submit your files regularly as you go! This will also prevent any late submissions. Additionally, compiling and testing your work on the teaching lab machines at intermediate stages will avoid last minute problems.

Submissions that are missing files or do not compile will receive a grade of 0. Submissions that do not execute (i.e., syntax errors), execute but do not produce an output (e.g., an empty file, a completely commented file), or have runtime errors, will all receive a grade of 0.

Further, all written solutions must be typed using LaTeX. In the event of diagrams, you may import a PDF or PNG image file into your main LaTeX document. All raw files must be submitted. Submissions that are not typed will receive a grade of 0.

Communications

It is the student's responsibility to:

- Maintain current contact information with the University, including address, phone numbers, and emergency contact information.
- Use the University provided e-mail address or maintain a valid forwarding e-mail address.
- Regularly check the official University communications channels. Official University communications are considered received if sent by postal mail, by fax, or by e-mail to the student's designated primary e-mail account via their @utoronto.ca or @mail.utoronto.ca alias.
- Accept that forwarded e-mails may be lost and that e-mail is considered received if sent via the student's @utoronto.ca or @mail.utoronto.ca alias.
- Check the UofT/Quercus email and course web site(s) on a regular basis during the term. Further, to check Quercus notifications daily for any course updates/content posting.
- Read and keep current with Piazza and the discussions therein throughout the term on a daily basis.

Academic Accommodations

Students who require academic accommodation must contact Accessibility Services (AS) to make arrangements with a Program Coordinator. Accessibility Services can be contacted by phone 905-569-4699 or e-mail access.utm@utoronto.ca. For further information, consult UofT's policy for Academic Accommodation: https://www.viceprovoststudents.utoronto.ca/student-resources/rights-responsibilities/academic-accommodation/. If a student with a disability chooses NOT to utilize an AS accommodation and chooses to write a standard midterm/exam, a petition for relief may not be filed retroactively (i.e. after the examination is complete).

Protection of Privacy Act (FIPPA)

The Freedom of Information and Protection of Privacy Act (FIPPA) applies to universities. Instructors should take care to protect student names, student numbers, grades and all other personal information at all times. For example, the submission and return of assignments and posting of grades must be done in a manner that ensures confidentiality. http://www.fippa.utoronto.ca/

Student Code of Conduct

The University of Toronto is committed to fostering a safe environment in which community members can pursue their educational goals. While the University assumes no general responsibility for the moral

and social behaviour of its students, there are cases in which the University's interest is unique and not adequately recognized by the wider justice system. For such instances, the University has its own set of internal offenses and procedures (found here: https://www.viceprovoststudents.utoronto.ca/student-resources/rights-responsibilities/). The Code of Student Conduct can be found here: https://governingcouncil.utoronto.ca/secretariat/policies/code-student-conduct-december-13-2019.

Creating a Positive Learning Environment

We are committed to creating a respectful learning environment in computer science courses for all students and expect that you will adhere to the University of Toronto Code of Student Conduct. Please be mindful of how your behaviour influences the atmosphere in our learning community, not just in classes, but also in labs, in online forums, and anywhere that you interact with other students and members of the department.

Specifically, we require all students to:

1. *Be welcoming*, friendly, and patient.
2. *Be respectful* in your speech and actions. Refrain from demeaning, discriminatory, harassing , or threatening speech and actions. This includes personal insults, discriminatory jokes, threats (or acts) of violence, bullying, unwanted sexual attention, or other repeated harassment.
3. *Seek to resolve conflicts constructively*. Disagreements and conflict are inevitable but should be handled patiently and respectfully. Conflict can be a source of learning if you seek to understand the source of the disagreement, and it can strengthen the community if resolved collaboratively.

NOTE: those who do not adhere to these rules/regulations, or violate the Student Code of Conduct, may be (temporarily or permanantly) banned from discussion boards (i.e., Piazza) and other course platforms if deemed necessary by the Course Instructor(s).

Crowdmark

We will be using Crowdmark for our Midterm/Test(s) and Final Exam, this is FREE and details will be posted on Piazza (the course disscusion board).

Use of Artificial Intelligence (AI)

- This course may use forms of artificial intelligence (AI) tools integrated in educational technologies (e.g., QuickTA, VoiceEx, etc.), when these tools are provided to by the teaching team, there will be specific instructions as to the boundaries and use-cases of these tools.
- In general, students may use artificial intelligence tools for learning course material, practicing programming, getting support coding, etc. However, all submitted work (i.e., any graded assessments) must be the original work produced by the individual student alone.
    - For example, submitting material that matches an original source without appropriate citation will be considered plagiarism, and submitting work with hallucinated sources (as

Large Language Models are known to do) will be considered fabrication. Both are academic offenses.

- Furthermore, the tests and exam in this course will not allow the use of AI tools (including generative AI) as aids, so you must be prepared to solve exercises, including writing Python, Java, and SQL code, without assistance. As a result, it is highly recommended that generative AI be used as a learning aid and that any material that is built on the output of generative AI tools be reviewed, so that you are confident that it reflects what you understand and can produce on your own.

- If you are not certain or have any question about the use of artificial intelligence applications in this course, please speak with the Course Instructor(s).

Equity, Diversity and Inclusion

The University of Toronto is committed to equity, human rights and respect for diversity. All members of the learning environment in this course should strive to create an atmosphere of mutual respect where all members of our community can express themselves, engage with each other, and respect one another's differences. U of T does not condone discrimination or harassment against any persons or communities.

Course Contacts

Course Website: https://q.utoronto.ca/courses/416678

Course Instructor: Prof. Dr. Michael Liut
Email: michael.liut@utoronto.ca
Office Hours and Location: Wednesdays from 5-7pm in DH-3074 (January 14th - April 1st, excluding Reading Week). Additional Office Hours may be scheduled by appointment.
Additional Notes: Emails will usually be returned within three (3) business days (i.e., 72-hours, not including weekends and school recognized holidays/breaks).

Course Overview

Introduction to database management systems. The relational data model. Relational algebra. Querying and updating databases: the query language SQL. Application programming with SQL. Integrity constraints, normal forms, and database design. Elements of database system technology: query processing, transaction management.

Course Learning Outcomes

Learning Outcomes

By the end of the course, **students should know and understand**:

1. The significance of the Entity-Relationship model and be able to list its main components.
2. When views, indexes, and constraints should be used, and describe the performance considerations associated with each construct.
3. The differences among indexing schemes, and when they are applicable.
4. The problems caused by redundancy in database design.
5. The 4 transactional properties that the DBMS guarantee.
6. Several data-intensive sustainability issues.
7. The different locking protocols used in a DBMS and their properties and behaviour.
8. Deadlock prevention and detection mechanisms.

By the end of the course, **students should be able to**:

1. Use a relational database management system to compose tables, load data into tables, and to write SQL to retrieve desired data records.
2. Write a query using relational algebra (RA), as well as, translate RA to SQL (and vice versa).
3. Develop an Entity-Relationship model for a given application. Translate a given Entity-Relationship model to a relational model.
4. Given a data instance, identify whether its normalized (e.g., 3NF, BCNF, etc.), and whether a decomposition is dependency preserving or a lossless join. If not, describe the normalization/decomposition steps.
5. Identify when anomalies can occur in interleaved transactions.
6. Identify a serializable schedule.
7. Understand the outcome of a series of transactions and any detect deadlocks that may result.

Prerequisites: CSC263H5
Corequisites: None
Exclusions: CSC343H1 or CSCC43H3
Recommended Preparation: None
Credit Value: 0.5

Course Materials

Textbooks and Other Materials

Required Textbook

- Database Management Systems - 3rd Edition. R. Ramakrishnan and J. Gehrke. McGraw-Hill. ISBN: 0072465638 // 978-0072465631.

Supplementary Textbooks

- Database Systems Concepts - 7th Edition. A. Silberschatz, H. F. Korth, and S. Sudarshan. McGraw-Hill. ISBN: 0078022150// 978-0078022159.

- Database Systems: The Complete Book. H. Garcia-Molina, J. D. Ullman, J. D. Widom. ISBN: 0130319953 // 978-0130319951.

Other Material

- Access to a computer*, stable/reliable internet, and an SSH client.

**despite UofT's Recommended Technology Requirements for Remote/Online Learning being the minimum technical requirements for this course, it is recommended that you adhere to UTM's ICCIT's Technical Requirements as they better align with minimal technical requirements for computing courses (including this one).*

Marking Scheme

Assessment	Percent	Details	Due Date
Assignment 1	10%	Due by 5:00PM.	2026-02-12
Assignment 2	10%	Due by 5:00PM.	2026-03-12
Assignment 3	10%	Due by 5:00PM.	2026-04-02
Weekly PCRS	10%	Weekly PCRS (videos, exercises, readings, reflections, external surveys, VoiceEx, QuickTA, external applications, etc.)	Ongoing
Midterm/Term Test	20%	Midterm will be held during lecture time.	2026-03-03, 2026-03-04
Final Assessment	40%	The Final Exam is cumulative. You must earn 40% or above on the Final Exam to pass the course; otherwise, your final course mark will be set no higher than 47%. Note: bonus points do NOT affect the 40% minimum requirement on the Final Exam.	Final Exam Period

Late Assessment Submissions Policy

NOTE: all dates and times reflect the time zone in which the University of Toronto Mississauga is in.

PCRS

PCRS contains weekly preparation video modules (sometimes readings) and exercises that are required to be completed prior to attending lecture. This will be recapped in a high-level at the beginning of

lecture. There will also be reflection components on these video modules, as well as on some of the assigned readings (both technical and non-technical) that will support your development in databases and computer science in general.

To receive full marks for PCRS, you must correctly complete all exercises, watch all the videos (or, if applicable, complete all readings), successfully complete all reflections (include follow-up components), complete all external surveys, and complete any other practice that is linked to external platforms. Exact deadlines and details will be provided on PCRS directly.

NOTE: *PCRS exercises must be completed* **individually***.*

Assignments

This course will have a total of **three (3) Assignments** worth a total of **30%** (each Assignment is worth 10%) of your final mark. Assignments must be **submitted by 5:00 PM** on the day due, unless otherwise communicated by the Course Instructor (check the course web site for details).

Assignments are to be **completed individually**. Assignments will be submitted on the MarkUs platform.

For accommodations, the Course Coordinator must approve this **in writing at least one week** *in advance* **of the due date**. To submit your Assignment, check your work into your git repository (on MarkUs) -- you may use the web interface. You **MUST understand ALL** parts of your Assignment. Teaching Staff (Instructors, Teaching Assistants, or a Course Coordinator Appointed Designate) may schedule and conduct interviews for some or all Assignments, after the due date, to determine if you have a good understanding of the material. Those who do NOT, may receive a reduced grade or a grade of 0 on the Assignment(s). Interview grades are final and cannot be appealed.

Please see the "Minimum Standards for Submitted Work" and the course web site for further details.

NOTE: *Assignments must be completed individually. Students are forbidden from collaborating with others (i.e., you cannot discuss your solutions/ideas with anyone else, however, you may seek assistance/clarification from the CSC343 Teaching Staff, but not other students, individuals, forums, external services, or external web sites). Please read our "Use of Artificial Intelligence (AI)" statement in detail, as all submitted work (i.e., any graded assessments) must be the original work produced by the individual student alone.*

Midterm/Test

This course has one in-person Midterm. The duration of this Midterm is 1-hour and 45-minutes and is worth a total of 20% of your final mark. It will occur in your lecture section on either **Tuesday, March 3rd, 2026 or Wednesday, March 4th, 2026** (in your registered lecture section). I encourage you to arrive early! Details will be posted on the course discussion board prior. If you are late by 30-minutes or more,

you will not be permitted to write the Midterm and will be awarded a grade of 0. Additional details will be provided via the course web site.

NOTE: *the Midterm is written in-person and must be completed individually.*

Final Exam

The Final Exam is worth 40% of your final grade and will test cumulative knowledge, with a heavier emphasis on material not covered by the Midterm/Test (concepts will build off of the Midterm/Test). You must earn 40% or above on the Final Exam to pass the course; otherwise, your final course mark will be set no higher than 47 (out of 100). Any bonus marks earned throughout the course does not change this.

The Final Exam will be scheduled by the Office of the Registrar at UTM. Details of the exam will be provided closer to the end of the semester. It is your responsibility to check the details, including the room location, of the exam as they are subject to change at any time.

Bonus/Extra Credit

There will *likely* be a few surveys/exercises that occur during the semester. Likely one at the beginning, one in the middle, and one at the end of the semester. In total, you *could earn* up to 3% in BONUS on your final grade by fully and **correctly completing** these surveys/exercises (we do not tolerate random guessing or button mashing, so we will check for this and not award a bonus to those who do randomly guess/button mash); details will be provided to you via the course discussion board. If these surveys/exercises do run in this course, they are optional, and as this bonus is not part of the normal 100% it will add to your overall grade in the course (up to a maximum final grade of 100%).

Penalties for Lateness

PCRS

PCRS modules/exercises, reflections, readings/videos, surveys, etc., must be done prior a student's lecture section (i.e., before a student's registered LEC section). Late submissions will not be accepted. Accomodations cannot be granted for these exercises as you need to have completed the materials to actively participate in the lectures.

Assignments

For Assignments, each student will receive **ten grace tokens total**; each grace token can be used for a **two-hour extension** for any Assignment. For example, you may choose to use all ten grace tokens on

Assignment 1, extending its deadline by twenty hours. Or, you may wish to use five tokens for Assignment 1 and Assignment 2, extending each deadline by ten hours.

It is your responsibility to submit your work on time. An assignment submitted at 5:00:01PM or after, are considered late and a grace token will be deducted automatically in MarkUs, if you have any remaining. We cannot give you back a grace token that MarkUs already deducted for work submitted close to the deadline yet after it had passed, even if it's just a few seconds past the deadline! **If you run out of grace tokens and attempt to submit late, your submission will not be accepted.** This means you will receive a grade of 0 for that submission.

Midterm/Test

Late submissions will not be accepted (i.e., if the Test is submitted late, even by 0.01 second, a grade of 0 will be awarded).

Religious Holidays

If a religious holiday will keep you from completing any assigned work, please let the Course Instructor know as soon as possible (but **no later than two weeks before the due date**), and we will work out a reasonable accommodation.

Procedures and Rules

Missed Term Work

To request special consideration, you must email supporting documentation to the Course Coordinator **at least one week in advance or within a week after** the occurence of a serious event. In case of illness, you must also declare your absence on Acorn.

Exact accommodations will be determined on a case-by-case basis and will not be given automatically. In other words, you risk getting a mark of zero for missed work unless you contact the Course Instructor promptly. *The Course Instructor reserves the right to replace any missed term work (assignments, PCRS, midterm/test, or otherwise) with a comprehensive oral examination or reweighting of the assessments; it will be up to the Course Instructor to choose which replacement method will be used and when it will be applied.*

Some general advice, if you have any concerns or questions regarding your situation, please contact your Department's Academic Counsellor or your Registrar's Office, as they are well-equipped to help you with anything you may be going through.

NOTE: you are **required to check Piazza, your UofT email, and the course web site (including download site) on a daily basis** for announcements, updates, content/material, etc., related to the course.

Policies & Statements

Academic Integrity

The Code of Behaviour on Academic Matters states that:

The University and its members have a responsibility to ensure that a climate that might encourage, or conditions that might enable, cheating, misrepresentation, or unfairness is not tolerated. To this end, all must acknowledge that seeking credit or other advantages by fraud or misrepresentation, or seeking to disadvantage others by disruptive behaviour, is unacceptable, as is any dishonesty or unfairness in dealing with the work or record of a student.

It is your responsibility as a student at the University of Toronto to familiarize yourself with, and adhere to, both the Code of Student Conduct and the Code of Behaviour on Academic Matters.
This means, first and foremost, that you should read them carefully.

Code of Student Conduct and the Code of Behaviour on Academic Matters are available from the U of T website.

Religious Accommodations

Information about the University's Policy on Scheduling of Classes and Examinations and Other Accommodations for Religious Observances is at https://www.viceprovoststudents.utoronto.ca/student-resources/rights-responsibilities/accommodation-religious/

Declaration of Temporary Absence

Students who miss an academic obligation during the term (i.e., in-class assessment, quiz, paper or lab report) may use the ACORN Absence Declaration Tool to record an absence in one or more courses. Students may utilize this option once per term for a single absence period of up to seven consecutive days. The declaration period must include the day of declaration and may include past and/or future dates, for a total of up to 7 calendar days.

Use of the ACORN Absence Declaration does not require supporting documentation and should be used in addition to the missed term work policy outlined in the course syllabus. It remains the student's responsibility to initiate the process for missed academic obligations by following the instructions in the course syllabus.

Re-grading Term Work

A student who believes that their written term work has been unfairly marked may ask the person who marked the work for re-evaluation. Students have up to one month from the date of return of an item of term work to inquire about the mark. If the student is not satisfied with this re-evaluation, they may

appeal to the instructor in charge of the course if the work was not marked by the instructor (e.g., was marked by a TA). Such re-marking may involve the entire piece of work and may raise or lower the mark. For more information on policies regarding re-marking of term work, please refer to Re-marking Pieces of Term Work in the Academic Calendar.

Plagiarism Software

A series of plagiarism detection software and tools (in addition to TurnItIn) will be utilized to detect the similarity of submissions in this course for the purpose of detecting plagiarism (e.g., MOSS, JPLAG, etc.). All submitted work in this course is subject to this verification.